

Архитектура ЭВМ и язык ассемблера

Семинар #37:

1. Устройство сопроцессора x87 FPU, базовая арифметика.
2. Сравнение чисел с плавающей точкой.
3. Передача чисел с плавающей точкой в функцию.
4. Сложная арифметика с x87 FPU.

Устройство сопроцессора x87 FPU, базовая арифметика



Состояние x87 FPU

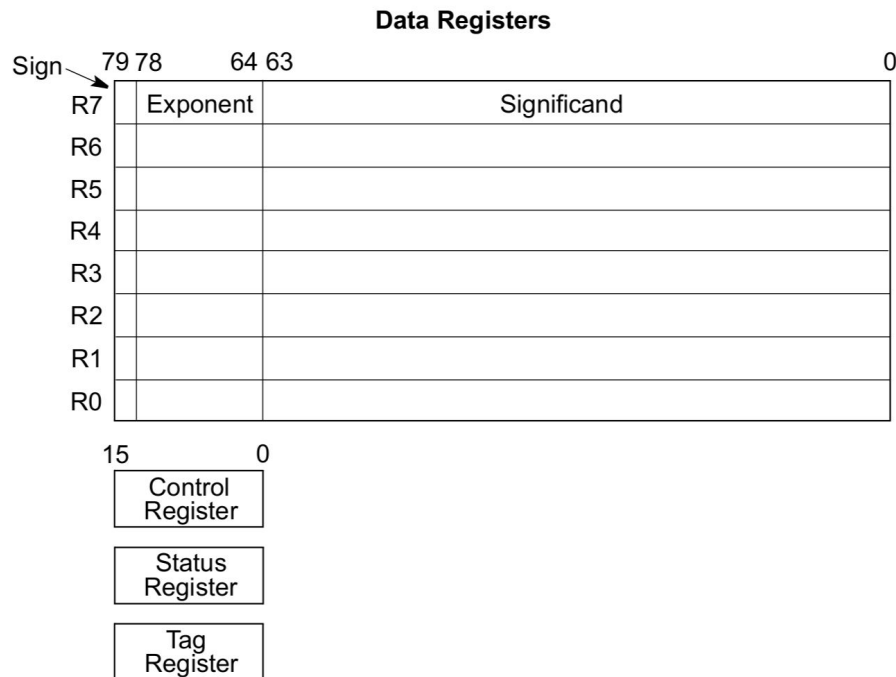


Figure 8-1. x87 FPU Execution Environment

Состояние x87 FPU

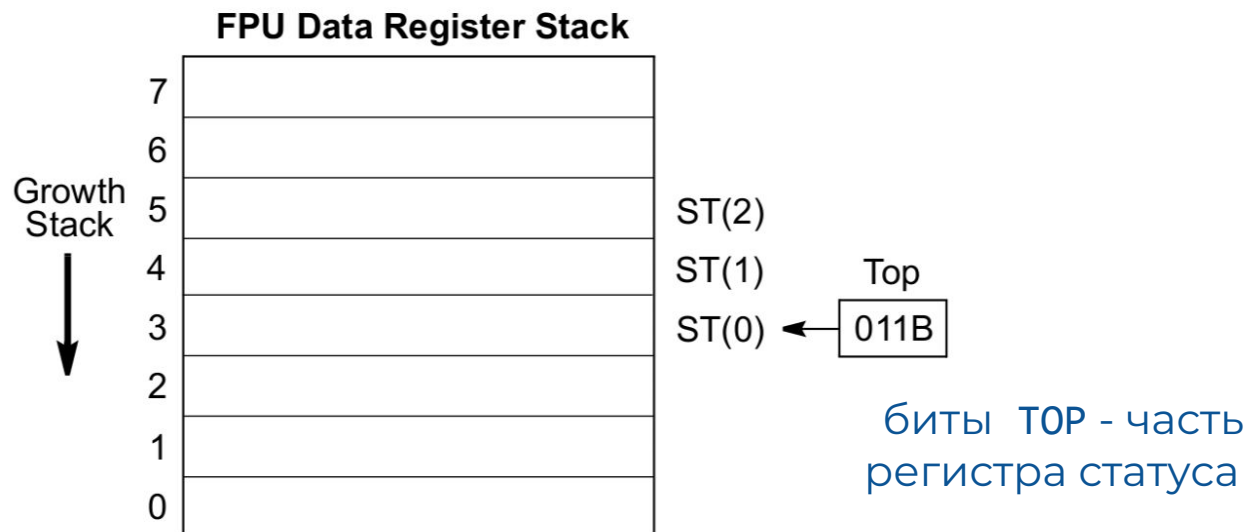


Figure 8-2. x87 FPU Data Register Stack

Инструкции x87 FPU

Table 8-4. Data Transfer Instructions

Floating-Point		Integer	
FLD	Load Floating-Point	FILD	Load Integer
FST	Store Floating-Point	FIST	Store Integer
FSTP	Store Floating-Point and Pop	FISTP	Store Integer and Pop
FXCH	Exchange Register Contents		
FCMOVcc	Conditional Move		

Базовая арифметика

Computation

Dot Product = $(5.6 \times 2.4) + (3.8 \times 10.3)$

Code:

```
FLD  value1 ; (a) value1 = 5.6
FMUL value2 ; (b) value2 = 2.4
FLD  value3 ; value3 = 3.8
FMUL value4 ; (c) value4 = 10.3
FADD ST(1)  ; (d)
```

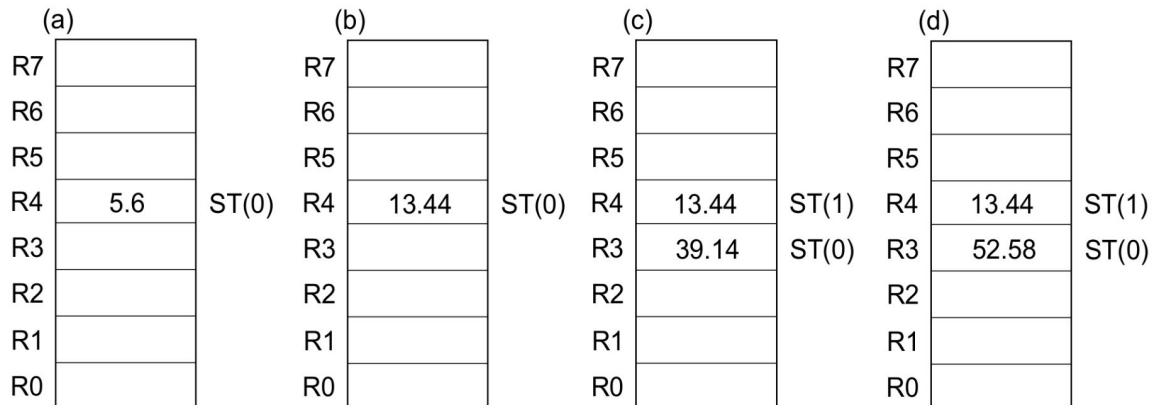


Figure 8-3. Example x87 FPU Dot Product Computation

Базовая арифметика

```
float f132_a = 1.0;  
float f132_b = 1.41421;  
  
f132_a = f132_b + 1.0;  
...
```

```
section .data  
sqrt2_dd: dd 0x3fb504d5
```

```
section .text  
push    ebp  
mov     ebp, esp  
sub     esp, 0x20  
fld1  
fstp    dword [ebp - 0x4], st0  
fld     st0, dword [sqrt2_dd]  
fstp    dword [ebp - 0x8], st0  
fld     st0, dword [ebp - 0x8]  
fld1  
faddp   st1, st0  
fstp    dword [ebp - 0x4], st0  
...
```

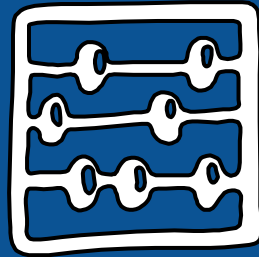
Базовая арифметика

```
...  
double f164_a = 1.0;  
double f164_b = 1.41421;  
  
f164_a = f164_b + 1.0;
```

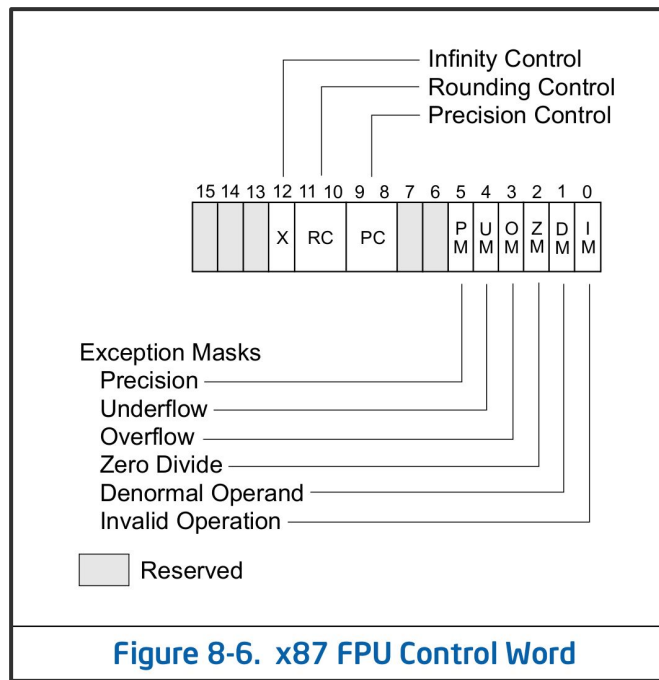
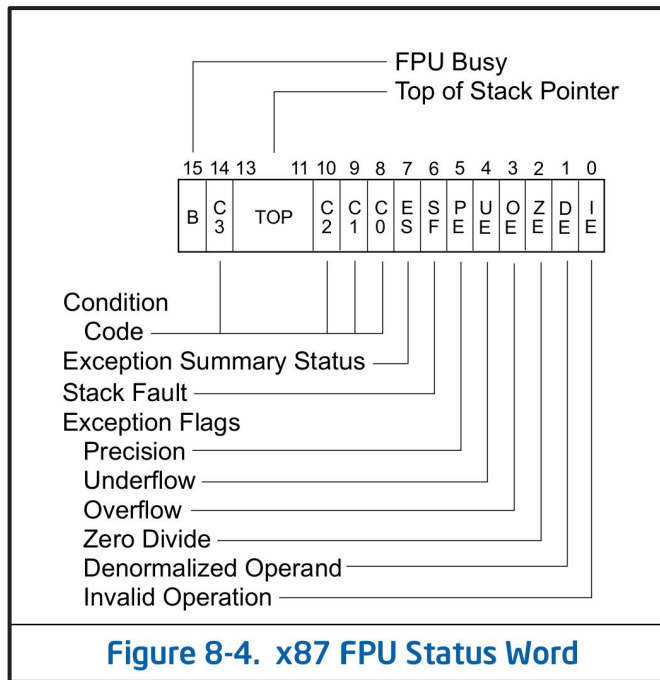
```
section .data  
sqrt2_dq: dq 0x3ff6a09aaa3ad18d
```

```
section .text  
...  
fldl  
fstp    qword [ebp - 0x10], st0  
fld     st0, qword [sqrt2_dq]  
fstp    qword [ebp - 0x18], st0  
fld     st0, qword [ebp - 0x18]  
fldl  
faddp   st1, st0  
fstp    dword [ebp - 0x10], st0
```

Сравнение чисел с плавающей точкой



Регистры флагов x87 FPU



У x87 FPU есть свои флаги (C0, C1, C2, C3),
но для сравнений он использует EFLAGS

Сравнение чисел при помощи x87 FPU

FCOMI ST, ST(i)	Compare ST(0) with ST(i) and set status flags accordingly.
FCOMIP ST, ST(i)	Compare ST(0) with ST(i), set status flags accordingly, and pop register stack.

Table 3-22. FCOMI/FCOMIP/ FUCOMI/FUCOMIP Results

Comparison Results*	ZF	PF	CF
ST0 > ST(i)	0	0	0
ST0 < ST(i)	0	0	1
ST0 = ST(i)	1	0	0
Unordered**	1	1	1

Сравнение чисел при помощи x87 FPU

```
float f132_a = 1.0;
float f132_b = 1.41421;

if (f132_a < f132_b)
{
    return 0;
}

return -1;
```

```
section .text
fld     st0, dword [ebp - 0x4]
fld     st0, dword [ebp - 0x8]
fcomip  st0, st1
fstp    st0, st0
jbe     .not_less
mov     eax, 0x0
jmp     .exit
.not_less:
mov     eax, 0xffffffff
.exit:
leave
ret
```

Условные перемещения x87 FPU

Table 8-5. Floating-Point Conditional Move Instructions

Instruction Mnemonic	Status Flag States	Condition Description
FCMOVB	CF=1	Below
FCMOVNB	CF=0	Not below
FCMOVE	ZF=1	Equal
FCMOVNE	ZF=0	Not equal

Table 8-5. Floating-Point Conditional Move Instructions (Contd.)

Instruction Mnemonic	Status Flag States	Condition Description
FCMOVBE	CF=1 or ZF=1	Below or equal
FCMOVNBE	CF=0 or ZF=0	Not below nor equal
FCMOVU	PF=1	Unordered
FCMOVNU	PF=0	Not unordered

Передача чисел с плавающей точкой в функцию



Обработка float в функции

```
int g(int a);

float f(float a, int b)
{
    return a * g(b);
}
```

Передача аргумента в ST(0)
Возвращаемое значение в ST(0)

```
f:
enter    8, 0
sub      esp, 0x4
push     dword [ebp + 0xC]
call     g
add      esp, 0x8
mov      dword [ebp - 0x4], eax
fild     st0, dword [ebp - 0x4]
fmul     st0, dword [ebp + 0x8]
leave
ret
```

Сложная арифметика с x87 FPU

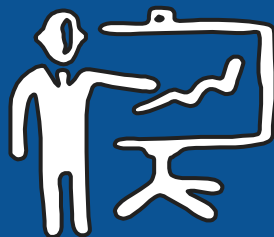


Инструкции x87 FPU

FADD/FADDP	Add floating-point.
FIADD	Add integer to floating-point.
FSUB/FSUBP	Subtract floating-point.
FISUB	Subtract integer from floating-point.
FSUBR/FSUBRP	Reverse subtract floating-point.
FISUBR	Reverse subtract floating-point from integer.
FMUL/FMULP	Multiply floating-point.
FIMUL	Multiply integer by floating-point.
FDIV/FDIVP	Divide floating-point.
FIDIV	Divide floating-point by integer.
FDIVR/FDIVRP	Reverse divide.
FIDIVR	Reverse divide integer by floating-point.
FABS	Absolute value.
FCHS	Change sign.

FLDZ	Load +0.0.
FLD1	Load +1.0.
FLDPI	Load π .
FLDL2T	Load $\log_2 10$.
FLDL2E	Load $\log_2 e$.
FLDLG2	Load $\log_{10} 2$.
FLDLN2	Load $\log_e 2$.

Вопросы?



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